

TechnoFunda Scanner & Analysis App

Product Specification, System Design & Incremental Development Roadmap

Development strategy: Build the Scanner + Stock Analysis product first, validate each component independently, then progressively add portfolio, monitoring, backtesting and AI capabilities.	
Document	Product specification and implementation roadmap
Methodology basis	TechnoFunda methodology document supplied by the user
Initial scope	Market Scanner + Stock Analysis
Later scope	Portfolio, alerts, backtesting, AI research layer
Design principles	Explainable, evidence-driven, modular, testable

Boundary: This is a research/analysis application specification. It does not create personalized investment advice or reproduce proprietary Shaktimaan internals.

1. Executive Summary

The supplied methodology describes a funnel that starts with price-volume behaviour, applies technical structure, performs fundamental screening and research, classifies businesses using growth and RoCE, and connects the research to position and exit processes. Its seven-step sequence is price-volume early warning → technical filter → fundamental screening → fundamental research → risk-based position sizing → early-warning exit signals → final technical/fundamental exit.

For the first release, the application should intentionally stop before portfolio execution. The first objective is: **Which stocks deserve deeper research, and what evidence supports the current state of each stock?**

Therefore V1 is two connected products: **Scanner** to reduce the market universe to candidates, and **Stock Analysis** to explain a candidate through price/volume, Stage Analysis, the four-cylinder earnings engine, MACHINE and the RoCE-growth framework.

1.1 Product vision

Market Universe → Candidate Detection → Technical Context → Fundamental Context → Business Research → Evidence-backed Thesis Workspace

Avoid a single opaque score initially. The methodology is better represented as a chain of independently inspectable observations. This makes the system easier to debug, test and later backtest.

1.2 Primary users

- Individual investor/researcher using the framework as a structured research process.
- Quant/ML engineer testing reproducible components.
- Future AI-assisted research workflow using source-backed company documents.

2. Methodology Translation into Software

Methodology component	Software module	Release
Price-volume action	Market scanner + price/volume engine	V1
Stage analysis	Technical structure engine	V1
Four cylinders	Earnings engine	V1
MACHINE	Business research workspace	V1
RoCE × earnings growth	Quality-growth classification	V1
Banyan / Bamboo	Classification and research tags	V1
Core / satellite allocation	Portfolio module	Later
Buy-add-sell / pyramiding	Portfolio + journal	Later
Exit monitoring	Alerts + monitoring	Later
Shaktimaan	Optional legitimate signal input only	Do not recreate
Backtesting	Historical strategy engine	Later
AI research	RAG / agent layer	Later

2.1 Four-cylinder engine

The methodology decomposes earnings growth into Sales Growth, Operating Leverage, Margin Expansion and Debt Reduction, and emphasizes identifying which cylinders are about to fire rather than only recording historical results.

Cylinder	Metrics / evidence	Possible interpretation
Sales Growth	3Y/5Y/10Y revenue CAGR; segment/geography growth; volume vs price where available; capex/capacity	Acceleration / stable / deterioration
Operating Leverage	Revenue growth vs EBITDA growth; incremental EBITDA margin; cost structure	EBITDA growing faster than revenue
Margin Expansion	Gross/EBITDA/EBIT/PAT margin trend; input-cost ratios; pricing evidence	Sustained margin improvement
Debt Reduction	Debt, D/E, net debt, interest coverage, CFO, FCF	Deleveraging + cash-flow improvement

2.2 MACHINE framework

Letter	Question in the app	Evidence examples
M	Is there a multi-year structural tailwind?	Industry growth, regulation, technology, import substitution
A	Is something actually happening now?	Capex, commissioning, product launch, regulation, management action
C	Is there a durable competitive advantage?	Brand, switching costs, network effects, scale, patents, cost position
H	Is the business high quality / high RoCE?	ROCE level, consistency, reinvestment ability
I	Are operating metrics improving?	Margins, working capital, ROCE, debt, returns
N	What is new?	Product, geography, management, segment, capital allocation
E	Does management execute?	Guidance vs delivery, historical execution

3. Scanner Specification

The scanner is the first major product. The methodology describes price-volume screens built from time-series data, with conditions involving price/volume relative to trailing averages, sector mapping and 52-week-high proximity. It names screen families such as large-consolidation breakouts, new-high breakouts, volume shakeouts and sectoral trends.

3.1 Architecture

Market Universe → Historical OHLCV → Feature Calculation → Signal Rules → Candidate Set → Analysis Page

3.2 V1 scanner families

Scanner	Core condition	Supporting fields
Large Consolidation Breakout	Price exits a defined consolidation range	Range duration, breakout level, volume, 52W distance
New High Breakout	Price reaches/breaks a defined 52-week high condition	Volume, trend, relative strength
Volume Expansion	Volume materially exceeds trailing baseline	Volume ratio, price change, sector
Price + Volume Surge	Unusual price movement with unusual volume	Trailing averages, sector, 52W position
Sector Trend	Sector relative performance improves	Sector return, benchmark-relative strength, constituent signals

3.3 Technical feature store

- Daily return and multi-day returns.
- 20/50/100/200-day moving averages.
- 30-week moving average or equivalent weekly long-term trend series.
- Volume moving averages and volume ratios.
- 52-week high/low distance.
- Rolling volatility.
- Breakout range high/low and consolidation duration.
- Relative strength versus selected benchmark.
- Sector-relative performance.
- Liquidity measures such as average traded value.

3.4 Scanner result contract

Field	Example
Symbol	ABC
Signal	Consolidation breakout
Breakout level	■XXX
Current price	■XXX
Volume ratio	2.4× 20D average
52W high distance	3.2%
Stage	Stage 2 candidate
Sector	Example sector

Field	Example
Signal timestamp	YYYY-MM-DD
Rule version	scanner_v1.3

3.5 Scanner UX

- Preset scanners for the documented signal families.
- Advanced filter builder with AND/OR conditions.
- Saved custom scanners.
- Sorting by signal freshness, volume ratio, breakout magnitude, relative strength, etc.
- Click any result to open Stock Analysis.
- Every result displays **why it triggered** and the actual calculation values.
- Every result records the rule version that generated it.

4. Stock Analysis Specification

The analysis page consolidates technical and fundamental evidence into a repeatable research template.

STOCK HEADER → PRICE / VOLUME + STAGE → FUNDAMENTAL SNAPSHOT → FOUR-CYLINDER ENGINE → MACHINE → RoCE × GROWTH → BANYAN/BAMBOO CONTEXT → RESEARCH NOTES / THESIS

4.1 Technical panel

Component	Display
Price chart	Daily and weekly chart
30-week MA	Weekly long-term trend reference
Volume	Volume bars + trailing average
Relative strength	Stock vs selected benchmark
Breakout	Range, breakout level, confirmation metrics
Stage	Stage 1/2/3/4 + supporting evidence
Trend state	Rising / flat / falling
Signal history	Historical scanner signals and stage transitions

4.2 Stage Analysis

V1 should implement a transparent approximation of the documented Stage framework. It should not treat price being above a moving average as sufficient for Stage 2.

Stage	V1 interpretation
Stage 1 — Basing	Sideways structure + flattening long-term MA + reduced deterioration; watch state
Stage 2 — Advancing	Breakout/advance + rising long-term MA + price structure + volume/relative-strength support
Stage 3 — Topping	Advance stalls + MA flattens + whipsaw/distribution characteristics
Stage 4 — Declining	Price below falling long-term MA + lower highs/lows + weak rallies

All thresholds must be configurable and versioned.

4.3 Fundamental panel

- Annual and quarterly revenue, EBITDA, EBIT, PAT and EPS.
- Gross, EBITDA, EBIT and PAT margins.
- Debt, cash and net debt.
- CFO, capex and FCF.
- ROCE and ROE.
- Shares outstanding and dilution.
- Promoter/institutional ownership where available.
- Segment/geography information where available.

- Allow configurable thresholds.
- Keep classification descriptive rather than making it an automatic trade instruction.

8. Technical Architecture

A Python-first architecture fits the planned application and keeps analytics reusable by the API, scheduled jobs and future backtester.

WEB UI → FastAPI → Market Data / Analytics / Research APIs → PostgreSQL + Feature Store → Scanner Engine → Signal Store

Layer	V1 choice	Purpose
Frontend	React / Next.js	Interactive scanner, charts and research UI
API	FastAPI	Python-native service layer
Database	PostgreSQL	Company, financial, signal and research data
Market data	Provider abstraction	Avoid coupling to one vendor
Analytics	Python + pandas/polars	Feature and signal calculations
Charts	Interactive chart library	Price/volume/stage visualization
Jobs	Cron/worker	Daily ingestion and recalculation
Cache	Redis optional	Fast repeated scanner/API queries
AI later	RAG + LangGraph	Source-backed research workflow

8.1 Core data entities

- Company — identity, sector, industry, exchange.
- PriceBar — date, OHLCV, adjusted fields.
- FinancialPeriod — revenue, EBITDA, EBIT, PAT, EPS, debt, cash, CFO, capex, etc.
- Feature — calculated metric + timestamp + feature version.
- Signal — scanner name, rule version, triggered values, timestamp.
- StageSnapshot — stage, evidence flags, rule version.
- MachineAssessment — letter, evidence, user assessment, confidence, review date.
- ResearchNote — thesis, catalyst, risk, invalidation, source references.
- Watchlist — user/company relationship.
- RuleVersion — exact scanner/technical/fundamental configuration.

9. API Specification — V1

Method	Endpoint	Purpose
GET	/companies	Search/filter companies
GET	/companies/{symbol}	Company overview
GET	/companies/{symbol}/prices	Historical price/volume
GET	/companies/{symbol}/fundamentals	Financial history
GET	/companies/{symbol}/technical	Technical features + stage
GET	/companies/{symbol}/earnings-engine	Four-cylinder analysis
GET	/companies/{symbol}/machine	MACHINE workspace

Method	Endpoint	Purpose
PUT	/companies/{symbol}/machine	Save MACHINE assessment
GET	/companies/{symbol}/matrix	RoCE-growth classification
GET	/scanners	List scanners
POST	/scanners/run	Run scanner with parameters
GET	/signals	Retrieve signals
GET	/watchlist	User watchlist
POST	/watchlist	Add company

10. Incremental Development Phases

Each phase should produce a usable artifact and a testable result before the next layer is added.

Phase	Scope	Exit criterion	Indicative effort
0 — Foundation	Repository, environments, PostgreSQL, migrations, logging, CI	Backend + database + health checks	1–3d
1 — Market Data	Provider abstraction, OHLCV, company metadata	Reliable historical dataset for small universe	3–7d
2 — Technical Features	MAs, volume ratios, 52W levels, relative strength, breakouts	Feature tables + tests	3–5d
3 — Scanner V1	Consolidation, new-high, volume and price-volume scanners	Usable scanner UI	4–7d
4 — Stock Analysis V1	Charts, technical features, scanner signal history	Candidate analysis page	4–7d
5 — Fundamental Data	Annual/quarterly financials + ratios	Fundamental dashboard	5–10d
6 — Four Cylinders	Translate financial history into four drivers	Earnings Engine panel	4–7d
7 — MACHINE	Qualitative evidence and assessments	Research workspace	3–6d
8 — RoCE-Growth	Matrix + historical movement	Classification view	2–4d
9 — Unified Funnel	Connect scanner to analysis workflow	Complete V1 research flow	3–5d
10 — Watchlist/Journal	Save candidates, thesis and review history	Personal research system	3–5d
11 — Backtesting	Point-in-time signal replay + metrics	Research-grade evaluator	7–14+d
12 — Monitoring	Stage/signal/fundamental change alerts	Continuous monitoring	5–10d
13 — AI Copilot	RAG + structured research generation	Evidence-backed AI analyst	7–14d

11. Detailed First Development Sprint

Phase 0 — Foundation

- 1 Create repository structure for frontend/backend/data.
- 2 Set up PostgreSQL and migrations.
- 3 Add environment/config management.
- 4 Add structured logging and error handling.
- 5 Add formatting/lint/unit-test CI.
- 6 Create Company table and health endpoint.

Phase 1 — Market Data

- 1 Start with a manageable universe rather than the entire market.
- 2 Define provider interface: `get_companies()`, `get_prices()`, `get_financials()`.
- 3 Store normalized data and source metadata.
- 4 Handle corporate actions and missing observations explicitly.
- 5 Record source and ingestion timestamps.
- 6 Create incremental daily ingestion.

- 7 Validate continuity, duplicates and impossible values.

Phase 2 — Technical Features

- 1 Calculate daily/weekly moving averages.
- 2 Calculate rolling volume averages and ratios.
- 3 Calculate 52-week high/low distance.
- 4 Build breakout/consolidation features.
- 5 Build benchmark-relative strength.
- 6 Build Stage Analysis evidence features.
- 7 Version calculations.

Phase 3 — Scanner V1

- 1 Implement each scanner as a pure function over feature data.
- 2 Create unit tests with synthetic price/volume sequences.
- 3 Move thresholds into configuration.
- 4 Persist signals with timestamp and rule version.
- 5 Build scanner UI.
- 6 Add 'why this appeared'.
- 7 Replay historically before adding more scanners.

12. Testing & Validation Strategy

The research system must be reproducible. A scanner result should be traceable to the exact input data, calculated features and rule version.

12.1 Unit tests

- Moving averages and rolling calculations.
- CAGR and incremental-margin calculations.
- Debt/cash-flow derived metrics.
- Breakout detection.
- Volume-ratio detection.
- Stage transition rules.
- RoCE-growth quadrant assignment.

12.2 Golden datasets

Maintain a small manually verified set of stocks/dates with expected feature values and expected signals. Run it on every rule change.

12.3 Historical replay

Replay scanner rules over historical data and inspect whether signals use only information available on the signal date.

12.4 Point-in-time discipline

- Financial results become available to the model only after their publication/availability date.
- Do not use today's company classification in historical simulations.
- Store effective dates and publication dates separately where possible.
- For serious backtesting, account for delisted/removed companies to reduce survivorship bias.
- Version thresholds and feature definitions.

12.5 Explainability requirement

If a stock triggered a volume breakout, the UI should expose the observed volume, reference average, ratio, date and rule version. Users should not need source-code access to understand a signal.

13. Post-V1 Expansion

13.1 Portfolio

- Core/satellite classification.
- Position ledger and tranche history.
- Buy/add/sell journal.
- Concentration and attribution.
- Winner/loser audit.

The methodology describes a core/satellite structure and tranching/pyramiding. These should be configurable research concepts, not hard-coded personal instructions.

13.2 Monitoring

- Stage 2 → 3 warning.
- Stage 4 transition.
- Volume deterioration.
- Breakout failure.
- Four-cylinder deterioration.
- Margin/debt/ROCE changes.
- Execution or thesis review alerts.

13.3 Backtesting

Capability	Requirement
Signal replay	Run exact historical scanner rules
Point-in-time fundamentals	Use only data known at each date
Transaction simulation	Configurable costs/slippage
Metrics	CAGR, drawdown, volatility, Sharpe/Sortino, turnover, hit rate
Attribution	Sector, scanner type, stage, cylinder configuration
Robustness	Different periods, universes and parameter sets
Out-of-sample	Separate development and evaluation periods

14. AI Research Layer — Later

Add AI after the deterministic scanner and analysis engine are reliable. Otherwise, an LLM can hide data-quality or rule-definition problems.

Financial Documents → Chunking → Retrieval → Evidence Extraction → MACHINE / Four-Cylinder Analysis → Research Report

14.1 AI capabilities

- Summarize annual reports and investor presentations.
- Extract capacity, product, geography and management-action evidence.
- Compare management guidance with subsequent outcomes.
- Draft MACHINE evidence with source citations.
- Explain changes in the four-cylinder engine.
- Identify contradictions between narrative and financial metrics.
- Generate research questions instead of unsupported conclusions.
- Maintain a source-backed company research timeline.

14.2 Future agent architecture

Agent	Responsibility
Orchestrator	Coordinate research workflow
Fundamental Agent	Financial history + four cylinders
Technical Agent	Price/volume + Stage Analysis

Agent	Responsibility
Sector Agent	Sector trend + relative strength
MACHINE Agent	Qualitative evidence extraction
Evidence Agent	Verify source/date and reduce unsupported claims
Report Agent	Assemble structured research report

15. Recommended UI Screens

Screen	Purpose	Priority
Dashboard	Market overview + recent signals	V1
Scanner	Find candidates	V1
Scanner Detail	Explain why each result triggered	V1
Stock Analysis	Complete company research page	V1
Technical	Chart + Stage + signals	V1
Fundamentals	Historical financial analysis	V1
Four Cylinders	Earnings-growth engine	V1
MACHINE	Qualitative research	V1
RoCE-Growth Matrix	Classification	V1
Watchlist	Track candidates	P1
Research Journal	Thesis/review history	P1
Portfolio	Core/satellite + positions	P2
Alerts	Monitoring	P2
Backtest	Historical evaluation	P2
AI Research	RAG/agentive analysis	P3

15.1 Scanner UI

SCANNERS → Presets / Custom Filters → Results Table → Signal Explanation → Open Stock Analysis

15.2 Stock Analysis UI

Stock Header → Technical/Stage → Fundamental Snapshot → Four Cylinders → MACHINE → RoCE/Growth → Banyan/Bamboo Context → Notes/Thesis

16. Engineering Principles

- 1 Explainability over opaque scoring.
- 2 Keep scanner, technical, fundamental and research engines independent.
- 3 Version rules, features and data timestamps.
- 4 Enforce point-in-time correctness.
- 5 Separate raw evidence, derived metrics and human interpretation.
- 6 Make thresholds configurable.
- 7 Require reproducibility: same data + same rule version = same result.
- 8 Deliver something usable at every phase.
- 9 Add AI after deterministic foundations.
- 10 Do not invent proprietary Shaktimaan internals.

17. V1 Acceptance Criteria

Area	Acceptance criterion
Data	Historical OHLCV and fundamental data load with source/timestamp metadata
Technical	Features calculate consistently and pass tests
Stage	Stage classification is transparent and evidence-backed
Scanner	Results are reproducible and versioned
Scanner UX	User can filter, sort and open analysis
Fundamentals	Historical trends and derived metrics are visible
Four Cylinders	All four drivers have evidence panels
MACHINE	All seven components can be reviewed and annotated
Matrix	RoCE-growth position is plotted using configurable thresholds
Research	Notes/thesis/invalidation fields can be saved
Quality	CI runs tests before merge
Boundary	No automatic trade execution or hidden recommendation logic

18. Definition of Done for the First Milestone

The first meaningful milestone is complete when an arbitrary stock can move from scanner → analysis page and the application can answer:

- 1 Why did the stock appear in the scanner?
- 2 What is happening to price and volume?
- 3 What Stage is it in, and what evidence supports the classification?
- 4 How have revenue, EBITDA, margins, debt and cash flow changed?
- 5 Which four earnings cylinders are improving or deteriorating?
- 6 What evidence exists for each MACHINE component?
- 7 Where is the company on the RoCE × earnings-growth matrix?
- 8 What research questions remain unanswered?
- 9 Can every displayed number/signal be traced to a source, calculation or explicit user assessment?

19. Recommended Build Order

Use this sequence to minimize rework:

```
1. Repository + PostgreSQL + FastAPI 2. Company universe + OHLCV ingestion 3. Technical feature engine 4.
Scanner V1 5. Scanner UI 6. Stock chart + Stage Analysis 7. Fundamental data + dashboard 8. Four-Cylinder
engine 9. MACHINE workspace 10. RoCE × Growth matrix 11. Unified Stock Analysis page 12. Watchlist +
Research Journal 13. Backtesting 14. Monitoring / Alerts 15. AI Research Copilot
```

20. Source Boundary

This specification is based primarily on the supplied 11-page TechnoFunda methodology document. The document describes price-volume action, fundamentals, sector rotation, the four-cylinder earnings model, MACHINE, the RoCE-growth grid, Stage Analysis, core/satellite allocation, buy-add-sell/pyramiding, exit discipline and the seven-step funnel.

The document explicitly states that Shaktimaan is proprietary, its construction is not disclosed, and no public backtest/parameter set was identified. This specification therefore does not invent or claim to reproduce its internals.

The document also notes that public worked examples are subject to survivorship selection and that there is no independently verified long-run risk-adjusted track record for the methodology as a whole. The future backtesting module should therefore test reproducible components independently and enforce point-in-time controls.

Primary source: The TechnoFunda Method — methodology report supplied in this conversation.